

RAJDEEP GILL

(431) 998-9785 | hi.rajdeepgill@gmail.com | github.com/Rajdeep-Gill

EDUCATION

Bachelor of Science in Computer Engineering (Co-op)

University of Manitoba

GPA: 4.45/4.5

Relevant Coursework: Data Structures and Algorithms, Systems Engineering Principles, Parallel Programming

May 2026

Winnipeg, MB

EXPERIENCE

Software Engineer Intern

Robinhood

- Incoming intern for the Summer 2025 term, working on the brokerage team.

May 2025 - Aug 2025

Toronto, ON

Undergraduate Research Student

Biosensors Research Lab - University of Manitoba

- Designed and trained a convolutional neural network to predict biological cell viability from 10,000+ diffraction pattern images, achieving 85% accuracy and demonstrating the potential for non-invasive cell viability assessment.
- Enhanced particle diffraction centroid detection using computer vision, improving success rate by 17% (from 75% to 92%) and providing more accurate data for the CNN model.
- Developed a Python-based data visualization tool with an interactive GUI to process, filter, and display diffraction patterns and model predictions, enabling researchers to analyze and interpret data efficiently.
- Designed and implemented a scalable SQL database to centralize 100,000+ optical diffraction records, replacing a manual file-sharing system.

May 2024 - Dec 2024

Winnipeg, MB

Undergraduate Research Student

Biosensors Research Lab - University of Manitoba

- Applied dimensionality reduction techniques (t-SNE, PCA) to analyze and cluster datasets, uncovering key correlations that informed future research directions and improved data interpretability.
- Optimized data analysis workflows by implementing parallel processing techniques, reducing processing time by 90%.
- Developed an oscilloscope interfacing program to automate data collection, improving speed and accuracy.
- Delivered weekly progress updates, communicating findings and incorporating feedback to refine research approaches.

May 2023 - Aug 2023

Winnipeg, MB

PROJECTS

Google Docs Clone | React, TypeScript, HTML/CSS, Next.js

- Developed a Google Docs clone with real-time collaboration, user authentication, and document sharing
- Utilized Convex to store and sync document data, allowing multiple users to edit the same document simultaneously
- Implemented organization-wide document sharing and user permissions to control access to documents

Financial Dashboard | React, TypeScript, HTML/CSS, Next.js, Hono.js

- Created an expense and income tracking dashboard with user authentication and real-time data visualization
- Implemented a RESTful API to handle user data and transactions, utilizing a PostgreSQL database for data storage

Deaf Curling Accessibility Aid | C, ESP-32, SolidWorks

- Developed a wireless communication system using ESP-32's, enabling deaf curlers to relay instructions to each other
- Designed a PCB and 3D-printed enclosure for the ESP-32's, ensuring durability and portability
- Implemented a vibration motor and LED system to provide haptic feedback and visual cues for the skip's instructions

PUBLICATIONS

B. Arzhang, E. Kovacs, J. Lee, **R. Gill**, E. Salimi, D. Thomson, and G. Bridges, "Machine Learning Classification of Microplastics by Integrating Optical and Dielectrophoresis Features," in *IEEE AP-S/URSI 2025*.

B. Arzhang, G. Bridges, **R. Gill**, J. Lee, E. Kovacs, E. Salimi, and D. Thomson, "Dielectric Analysis of Single Biological Cells at Both Microwave and Optical Frequencies," in *IEEE AP-S/URSI 2025*.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML/CSS, R, SQL, MATLAB

Frameworks & Libraries: React, Next.js, TensorFlow, PyTorch, OpenCV,

Developer Tools: Git, VSCode, Visual Studio, Android Studio, IntelliJ